

Human SNP Links Differential Outcomes in Inflammatory and Infectious Disease to a FOXO3-Regulated Pathway

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Evidence abstract

The clinical course and eventual outcome, or prognosis, of complex diseases varies enormously between affected individuals.

This variability critically determines the impact a disease has on a patient's life but is very poorly understood.

Here, we exploit existing genome-wide association study data to gain insight into the role of genetics in prognosis.

We identify a noncoding polymorphism in FOXO3A (rs12212067: T > G) at which the minor (G) allele, despite not being associated with disease susceptibility, is associated with a milder course of Crohn's disease and rheumatoid arthritis and with increased risk of severe malaria.

Minor allele carriage is shown to limit inflammatory responses in monocytes via a FOXO3-driven pathway, which through TGF 1 reduces production of proinflammatory cytokines, including TNF , and increases production of anti-inflammatory cytokines, including IL-10.

Thus, we uncover a shared genetic contribution to prognosis in distinct diseases that operates via a FOXO3-driven pathway modulating inflammatory responses.

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Evidence checklist

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Source continuity

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